

ASE WEEK 02 — UNIT 07

The Unified Process

Putting it all together — four business-driven phases, nine workflows, and the one model that treats deployment as part of the process

Sources. Mall, *Fundamentals of Software Engineering* 4th ed., §8.3.1, pp.475–476 (primary — the phases and the use-case-driven framing). Sommerville, *Software Engineering* 9th ed., §2.4, pp.67–70 (hybrid framing, six best practices, workflows, three perspectives). **Note on method.** This is a compendium, not a summary. Every definition is quoted verbatim from the source before it is explained, and every claim carries a page anchor. Mall and Sommerville weight the phases differently — **both accounts are shown side by side and never merged.**

Where this sits

Unit Context: Unit 06 (Spiral) closed on a limitation of its own — the spiral is a **powerful way of thinking about risk**, but it is *not a ready-made framework* a team can adopt on Monday morning. It tells you to attack risk each loop; it does not package the work into named phases, practices, and workflows.

The question this unit answers: *can the good ideas from every earlier model be combined into one adoptable framework?* The Unified Process (UP, or RUP in its Rational flavour) is the attempt to do exactly that — it is, in Sommerville's words, **a hybrid model** that "brings together elements from all of the generic process models." [Foundational Knowledge / Standard Concept]

The question it hands to the next unit: *UP is disciplined and complete — and it is heavy. For a small team whose requirements move fast, even that is too much process — which is what forces the change of philosophy in the next unit.*

The connective tissue: UP reuses, in one package — prototyping (inception), iterative risk-reduction (elaboration ≈ spiral), incremental delivery (construction), and deployment-as-part-of-process (transition). It is the "put it all together" chapter the narrative spine promised.

1. What the Unified Process Is

Verbatim (Mall p.475):

Unified process is incremental in iterative process model for object-oriented software development that has gained acceptance among the practitioners and academicians. The first book to describe the unified process was titled "The Unified Software Development Process" and was published in 1999 by Ivar Jacobson, Grady Booch and James Rumbaugh.

Verbatim (Mall p.475):

The two main characteristics of the unified process are: **use case-driven** and **iterative**.

تعريفان أساسيان:

1. **موجه بـ use cases** — الـ use cases (وجهة نظر العميل) هي العرض المركزي والأهم. كل النماذج اللاحقة لازم تتوافق مع نموذج الـ use case.

2. **تكراري (iterative)** — يتطور عبر تكرارات قصيرة مودعة واحدة.

Verbatim (Mall p.475):

The use case model is the central model. All models that are constructed in the subsequent design activities must conform to the use case model.

Mall يصرح: نموذج الـ use case هو المركزي — وكل نماذج التصميم (class diagram, interaction, الخ) لازم تلتزم بيه. هذا يفرق UP عن باقي النماذج اللي تبدأ بالمطلوبات النصية (SRS).

Verbatim (Sommerville p.67):

The Rational Unified Process (RUP) (Krutchen, 2003) is an example of a modern process model... it is a good example of a **hybrid process model**. It brings together elements from all of the generic process models (Section 2.1), illustrates good practice in specification and design (Section 2.2) and supports prototyping and incremental delivery (Section 2.3).

Sommerville يؤكد طبيعة UP كنموذج هجين — يجمع عناصر من كل النماذج العامة. هذي هي "put it all together" بالمعنى الحرفي.

ملاحظة تغطية: Mall does not use the word "hybrid"; Mall frames UP by its "two that framing is Sommerville's. Both are correct cuts (iterative ,use-case-driven) characteristics recorded so a student does — and its OO focus. contradictions not read them as [Source Difference]

UP vs RUP — the naming

Verbatim (Mall p.475):

authors affiliated with Rational Software Corporation have favoured the name rational unified process (RUP).

UP = الاسم العام؛ **RUP** = النسخة المرتبطة بالـ Software Rational. الامتحان قد يذكر أحدهما — كلاهما نفس النموذج.

2. The Four Phases — Mall and Sommerville Side by Side

Per the agreed plan, the two accounts are shown side by side and NOT merged. Both books name the same four phases; they weight the *content* of each differently. [Source Difference]

Verbatim (Sommerville p.67):

The RUP is a phased model that identifies four discrete phases in the software process. However, unlike the waterfall model where phases are equated with process activities, the phases in the RUP are more closely related to **business rather than technical concerns**.

نقطة Sommerville الجوهرية: مراحل UP متعلقة بالأعمال (business) **مو بالتقنية** — عكس Waterfall اللي يربط كل مرحلة بنشاط تقني. وهذا يعني **المرحلة ≠ النشاط** — نفس النشاط (مثل المتطلبات) ممكن يصير بأكثر من مرحلة.

2.1 Inception

Mall p.476	Sommerville p.67
"the scope of the project is defined and prototypes may be developed to form a clear idea about the project."	"The goal of the inception phase is to establish a business case for the system. You should identify all external entities (people and systems) that will interact with the system and define these interactions... assess the contribution that the system makes to the business. If this contribution is minor, then the project may be cancelled after this phase."

Mall يرکز على **النطاق + النماذج الأولية**؛ Sommerville يرکز على **الحالة التجارية (business case)** — ولهذا قد يلغى المشروع إذا كان مساهمته ضئيلة. التكامل: النموذج الأولي يساعد بتوضيح النطاق اللي بدوره يبرر الحالة التجارية.

2.2 Elaboration — THE FLAGGED DISAGREEMENT (shown, not merged)

Verbatim (Mall p.476):

In this phase, the functional and the non-functional requirements are captured. **The preliminary use case and the domain model are developed** during this phase.

Verbatim (Sommerville p.68):

The goals of the elaboration phase are to **develop an understanding of the problem domain, establish an architectural framework for the system, develop the project plan, and identify key project risks**. On completion of this phase you should have a requirements model for the system, which may be a set of **UML use-cases**, an architectural description, and a development plan for the software.

هذا هو الاختلاف المعلم بالخطة — والاثنان **مو متناقضين**، بل يشدان على جوانب مختلفة:

- **Mall:** المطلوبات (وظيفية + غير وظيفية) + **use case** مبدئي + **domain model**.
- **Sommerville:** فهم مجال المشكلة + الإطار المعماري + خطة المشروع + المخاطر الرئيسية + (عند الإنجاز) نموذج مطلوبات (UML use-cases) + وصف معماري + خطة تطوير.

القاعدة: كلاهما يذكر **use cases**، بس Sommerville يضيف **الإطار المعماري والمخاطر** (يربط بالوحدة 106!)، بينما Mall يضيف **الmodel domain** (يربط بالتحليل والتوصيف الكائني OOAD). لو السؤال غال «حسب Mall» — جاب بمطلوبات + use case + domain model. لو غال «حسب Sommerville» — جاب بمجال + إطار معماري + خطة + مخاطر. [Source Difference]

2.3 Construction

Mall p.476	Sommerville p.68
"the design and implementation activities are carried out. Full text descriptions of use cases are written and each use case is taken up for the start of a new iteration. System features are implemented in a series of short iterations and are tested. Each iteration results in an executable release of the software."	"The construction phase involves system design, programming, and testing. Parts of the system are developed in parallel and integrated during this phase. On completion of this phase, you should have a working software system and associated documentation that is ready for delivery to users."

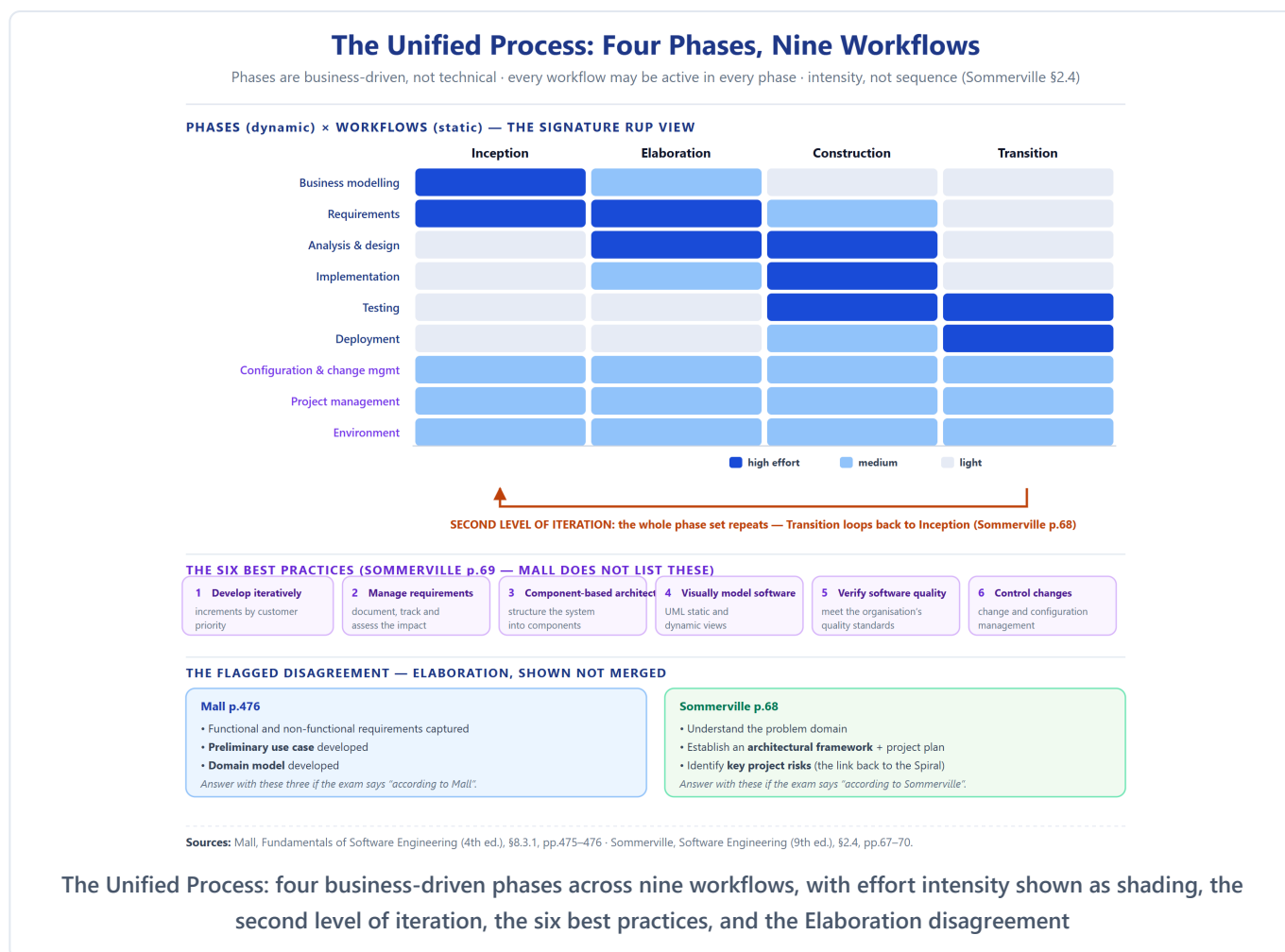
اتفاق واسع: تصميم + برمجة + اختبار + إصدار تنفيذي (executable release) بكل تكرار. Mall يشدّ على الـ **case use** الكامل النصي كوحدة تكرار؛ Sommerville يشدّ على التطوير المتوازي + الدمج.

2.4 Transition

Mall p.476	Sommerville p.68
"the product is installed in the user's environment and maintained. "	"The final phase of the RUP is concerned with moving the system from the development community to the user community and making it work in a real environment. This is something that is ignored in most software process models but is, in fact, an expensive and sometimes problematic activity. On completion... a documented software system that is working correctly in its operational environment."

اتفاق: النقل للبيئة الحقيقية + الصيانة. Sommerville يضيف نقطة مهمة: النشر متجاهل بأغلب النماذج لكنه مكلف ومشكلة أحياناً — وهذا «الابتكار» الأساسي بـUP (انظر §5).

2.5 The whole model in one picture



كيف تقرأ الرسم — وهذا يجاب على أهم سؤال بالوحدة:

العنصر في الرسم	معناه الهندسي
الأعمدة الأربعة	المراحل (dynamic view) — Inception · Elaboration · Construction · Transition
الصفوف التسعة	الworkflows (static view) — ستة أساسية + ثلاثة داعمة (Sommerville p.69)
تدرج اللون	شدة الجهد — أزرق غامق = جهد عالي، فاتح = قليل. مو تسلسل — كل workflows ممكن تكون نشطة بكل مرحلة
الشكل العام (المنحنى)	الجهد ينتقل من الأعمال والمطلوبات (يسار) إلى الاختبار والنشر (يمين)
السهم البرتقالي السفلي	التكرار بمستويين — المستوى الثاني: المراحل كلها تتكرر (Inception → Transition)
الشرائح البنفسجية الست	أفضل الممارسات الست — كلها من Sommerville؛ Mall ما يذكرهن
الصندوقان السفليان	الاختلاف المعلم بـ Elaboration — Mall مقابل Sommerville. يُعرضان معاً، ما يُدمجان

الخلاصة البصرية: UP مو خط مراحل — هو شبكة: المراحل بالعرض، والأنشطة بالطول. ولهذا Sommerville يسمي أهم ابتكار فيه «فصل المراحل عن workflows» (separation of phases and workflows). والابتكار الثاني: النشر جزء من العملية، مو

شي يصير بعدها.

3. Two-Level Iteration (Sommerville-only depth)

Verbatim (Sommerville p.68):

Iteration within the RUP is supported in two ways. Each phase may be enacted in an iterative way with the results developed incrementally. In addition, the whole set of phases may also be enacted incrementally, as shown by the looping arrow from Transition to Inception in Figure 2.12.

تكرار بمستويين:

1. داخل كل مرحلة — المرحلة نفسها تُنفَّذ تكررًا.

2. عبر كامل المجموعة — المراحل كلها تتكرر (Inception → Transition)، يعني دورة كاملة جديدة بعد التسليم.

ملاحظة تغطية: Mall describes construction as "a series of short iterations" but does not describe the "whole-set" iteration (looping back from Transition to Inception). That second level is Sommerville-only. [Source Difference]

4. The Three Perspectives (Sommerville-only)

Verbatim (Sommerville p.67):

The RUP is normally described from three perspectives: 1. A **dynamic perspective**, which shows the phases of the model over time. 2. A **static perspective**, which shows the process activities that are enacted. 3. A **practice perspective**, which suggests good practices to be used during the process.

ثلاث زوايا لفهم UP:

- ديناميكية = المراحل عبر الزمن (Inception → Elaboration → Construction → Transition).
- ساكنة = الأنشطة التقنية (workflows) — تشتغل بكل المراحل.
- ممارسة = أفضل الممارسات (الست اللي بالقسم 5).

ملاحظة تغطية: Mall presents UP only through the dynamic/phase lens (and the use-case lens); the three-perspective decomposition is Sommerville's. Recorded so a student knows Mall alone will not supply perspectives. [Source Difference]

5. The Six Best Practices (Sommerville-only)

Verbatim (Sommerville p.69):

Six fundamental best practices are recommended:

1. **Develop software iteratively** — Plan increments of the system based on customer priorities and develop the highest-priority system features early in the development process.
2. **Manage requirements** — Explicitly document the customer's requirements and keep track of changes to these requirements. Analyze the impact of changes on the system before accepting them.
3. **Use component-based architectures** — Structure the system architecture into components, as discussed earlier in this chapter.
4. **Visually model software** — Use graphical UML models to present static and dynamic views of the software.
5. **Verify software quality** — Ensure that the software meets the organizational quality standards.
6. **Control changes to software** — Manage changes to the software using a change management system and configuration management procedures and tools.

الست ممارسات — كلها من **Sommerville**: Mall ما يذكرها. وهي تلخص فلسفة UP:

1. تطوير تكراري — حسب أولوية العميل.
2. إدارة المطلوبات — توثيق + تتبع التغيير + تحليل الأثر.
3. معمارية قائمة على المكونات (component-based).
4. نمذجة بصرية بـ UML.
5. التحقق من الجودة.
6. ضبط التغييرات (configuration management).

ملاحظة تغطية: name the six best practices of" list. If an exam asks "best practices" Mall's account of UP has no [Source Difference] the answer is Sommerville's "RUP,

6. The Static Workflows (Sommerville-only)

Verbatim (Sommerville p.69):

The core engineering and support workflows are described in Figure 2.13. [Six core process workflows:] Business modelling, Requirements, Analysis and design, Implementation, Testing, Deployment. [Three supporting workflows:] Configuration and change management, Project management, Environment.

الworkflows (الأنشطة الساكنة) — ستة أساسية + ثلاثة داعمة. النقطة المهمة (70–Sommerville p.69):

Verbatim (Sommerville p.70):

The advantage in presenting dynamic and static views is that **phases of the development process are not associated with specific workflows**. In principle at least, all of the RUP workflows may be active at all stages of the process. In the early phases of the process, most effort will probably be spent on workflows such as business modelling and requirements and, in the later phases, in testing and deployment.

المرحلة ≠ workflow. كل workflows ممكن تكون نشطة بكل المراحل — بس التركيز يختلف (بداية = مطلوبات، نهاية = اختبار + نشر). هذي هي «separation of phases and workflows» اللي Sommerville يسميها ابتكار UP الأساسي.

ملاحظة تغطية: Mall does not describe workflows at all ;this is entirely Sommerville [Source Difference]

7. The Two Key Innovations — and When UP Does NOT Fit

Verbatim (Sommerville p.70):

The most important innovations in the RUP are the **separation of phases and workflows**, and the **recognition that deploying software in a user's environment is part of the process**.

ابتكاران: (1) فصل المراحل عن الأنشطة؛ (2) اعتبار النشر جزء من العملية (مو شي خارجها). هذا يجاوب على قصور كل النماذج السابقة اللي كانت تنهي عند التسليم.

Verbatim (Sommerville p.70):

The RUP is not a suitable process for all types of development, e.g., **embedded software development**. However, it does represent an approach that potentially combines the three generic process models discussed in Section 2.1.

الضعف/الحد: UP مو مناسب لكل شي — مثال Sommerville: الأنظمة المضمّنة (embedded). و"يجمع النماذج العامة الثلاثة" Reuse-oriented + Incremental + Waterfall = (من 2.1).

ملاحظة تغطية: a Mall does not state "when not suitable" limitation for UP the embedded-software caveat is ;Sommerville-only [Source Difference]

8. Source notes

Source	Role	What it adds	What it omits
Mall p.475–476	Primary	UP definition (OO, use-case-driven, iterative); the four phases with Mall's wording; the domain-model emphasis in Elaboration; UP vs RUP naming	No "hybrid" framing; no six best practices; no workflows; no "when not suitable"
Sommerville p.67–70	Supporting	Hybrid framing; business-not-technical phase nature; Elaboration with architecture+risks ; two-level iteration; three perspectives; six best practices ; static workflows ; embedded-software limitation	No domain-model emphasis in Elaboration

The decision rule applied (per student instruction): all sources were read in full before writing. Where Mall and Sommerville describe the four phases, **both accounts are shown side by side** (§2) and the Elaboration difference is presented explicitly (§2.2), not merged. Where Sommerville alone supplies material (six best practices, workflows, three perspectives, the limitation), the coverage note says so explicitly, so a student knows Mall will not supply it. Rule 3 (show both, never merge) and Rule 6 (a coverage note flags) honoured. [Source Difference]

9. Retrieval set

RS-07-01 What is the Unified Process, and who originated it?

An incremental and iterative process model for object-oriented software development. First described in *The Unified Software Development Process* (1999) by Jacobson, Booch, and Rumbaugh. "RUP" is the Rational-flavoured name. [Mall p.475]

RS-07-02 What are the two main characteristics of the UP?

Use-case-driven and **iterative**. The use-case model is the central model; all other models must conform to it. [Mall p.475]

RS-07-03 Why is the UP called a "hybrid process model"?

It brings together elements from all the generic process models (waterfall, incremental, reuse-oriented) and supports prototyping and incremental delivery. [Sommerville p.67]

RS-07-04 Name the four phases of the UP.

Inception, Elaboration, Construction, Transition. [Mall p.476; Sommerville p.67–68]

RS-07-05 How does the UP differ from the waterfall in how phases relate to activities?

In the waterfall, phases are equated with process activities; in the UP, phases are related to **business** rather than technical concerns, and the same workflow (e.g., requirements) can be active in multiple phases. [Sommerville p.67]

RS-07-06 What happens in the Inception phase, per Mall vs Sommerville?

Mall: scope defined, prototypes may be built. Sommerville: establish the **business case**; project may be **cancelled** if its business contribution is minor. [Mall p.476; Sommerville p.67]

RS-07-07 Describe Elaboration — and note the source difference.

Mall: functional + non-functional requirements captured; preliminary use case + domain model developed.

Sommerville: understand problem domain, establish architectural framework, develop project plan, identify key project risks; deliver a requirements model (UML use-cases), architectural description, development plan. Both mention use cases; they weight different facets. Do not merge. [Mall p.476; Sommerville p.68; Source Difference]

RS-07-08 What happens in Construction?

Design + implementation + testing; features implemented in short iterations, each yielding an executable release. Sommerville adds: parts developed in parallel and integrated. [Mall p.476; Sommerville p.68]

RS-07-09 What happens in Transition, and why does Sommerville call it an innovation?

Product installed in the user's environment and maintained. Sommerville: moving the system to the user community is "ignored in most models" yet "expensive and sometimes problematic" — so recognising deployment as *part of the process* is a key UP innovation. [Mall p.476; Sommerville p.68, p.70]

RS-07-10 What are the two levels of iteration in the RUP?

(1) Each phase enacted iteratively; (2) the whole set of phases enacted incrementally, looping from Transition back to Inception. [Sommerville p.68]

RS-07-11 What are the three perspectives from which the RUP is described?

Dynamic (phases over time), Static (activities/workflows enacted), Practice (good practices). [Sommerville p.67]

RS-07-12 List the six best practices of the RUP.

(1) Develop iteratively; (2) Manage requirements; (3) Use component-based architectures; (4) Visually model software (UML); (5) Verify software quality; (6) Control changes to software. [Sommerville p.69]

RS-07-13 What are the RUP workflows (static view)?

Six core: Business modelling, Requirements, Analysis and design, Implementation, Testing, Deployment. Three supporting: Configuration and change management, Project management, Environment. [Sommerville p.69]

RS-07-14 What is the key innovation of the RUP, according to Sommerville?

The **separation of phases and workflows**, and the **recognition that deploying software is part of the process**. [Sommerville p.70]

RS-07-15 When is the RUP NOT suitable?

Not for all development types — e.g., **embedded software development**. [Sommerville p.70 — Mall does not state this]

RS-07-16 If an exam says "according to Mall" vs "according to Sommerville" for the UP, what differs?

Mall = use-case-driven + iterative, OO focus, four phases with domain-model emphasis, no best-practices/workflows list. Sommerville = hybrid framing, business-not-technical phases, Elaboration with architecture+risks, six best practices, static workflows, embedded limitation. Show both; never merge. [Source Difference]

10. Page anchors

Mall: p.475 (UP definition, OO, use-case-driven + iterative, UP vs RUP, use-case model central) · p.476 (four phases: inception/prototypes, elaboration/requirements+use case+domain model, construction/short iterations+executable release, transition/install+maintain)

Sommerville: p.67 (RUP hybrid, four phases business-not-technical, three perspectives, inception business case) · p.68 (elaboration architecture+risks, construction parallel+integrated, transition deployment, two-level iteration) · p.69 (six best practices, static workflows table) · p.70 (key points, two innovations, embedded limitation)

End of Unit 07.

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